

Introduction

In this report we have put forward a Local Area Network (LAN) & Wide Area Network (WAN) design proposal for ABC Logistics, a transport and warehousing organization that is expanding rapidly and operates along the Australian East Coast. The company's main office is in Sydney. They also have branch offices in Brisbane, Canberra, and Adelaide. Because of the company's expansion, a network infrastructure that is secure, scalable, and reliable has been identified as one of the supports for communication, data sharing, cloud access and other business operations of the company in all its offices.

Efficient communication and performance of operations are very much dependent on the well-engineered LAN and WAN layouts in the present-day enterprise. To share resources like files, printers' servers, and internet connectivity, among other devices within a single location, LAN is used, whereas a WAN connects several offices located in different geographical locations and forms one network. For ABC Logistics, the network should be capable of supporting wired and wireless users, voice and video communications based on IP, cloud services, handheld devices in warehouses, and mobile staff going online through 4G/5G network.

Reliability and performance are two highly critical aspects of business networks, as network going down could disturb communication and could cause delays in logistics operations, and decrease productivity. A trustworthy network will guarantee a seamless link between different offices, while an efficient infrastructure would increase the rate at which files get transferred, the responsiveness of applications, the quality of voice and video communications, etc. Of course, as well as these two fundamental factors, the choice of communication media such as fiber optic cabling, Ethernet, and wireless technologies should not be overlooked to keep interference to a minimum, increase bandwidth, and provide support to future extensions.

Ensuring device compatibility is also a key thing when developing enterprise networks. It is necessary to have all the networking components, including routers switches wireless access points on server's printers, and mobile devices, operating together efficiently and securely. That's why the network plan presented to ABC Logistics mainly revolves around scalability security performance, and compatibility to cater not only to the present business operations but also to the growth of the organization in the near future.

LAN Network Design for Sydney Headquarters

The Sydney headquarters will use a hierarchical star topology LAN design. The network will consist of a central core switch located in the IT server room on Level 2, connected to access switches on

each floor. Wireless access points, printers, workstations, servers, routers, and firewalls will also be connected through this infrastructure.

Proposed Network Components

Component	Purpose
Core Layer 2 Switch	Central network management and VLAN routing
Access Switches	Connect workstations and printers
Wireless Access Points	Provide Wi-Fi coverage
Router	WAN and internet connectivity
Firewall	Network security and traffic filtering
Servers	File sharing, authentication, applications
Printers	Shared departmental printing
Wireless Devices	Tablets and warehouse scanners

SYDNEY HEAD OFFICE LAN DIAGRAM

- Network Diagram for Sydney Head Office

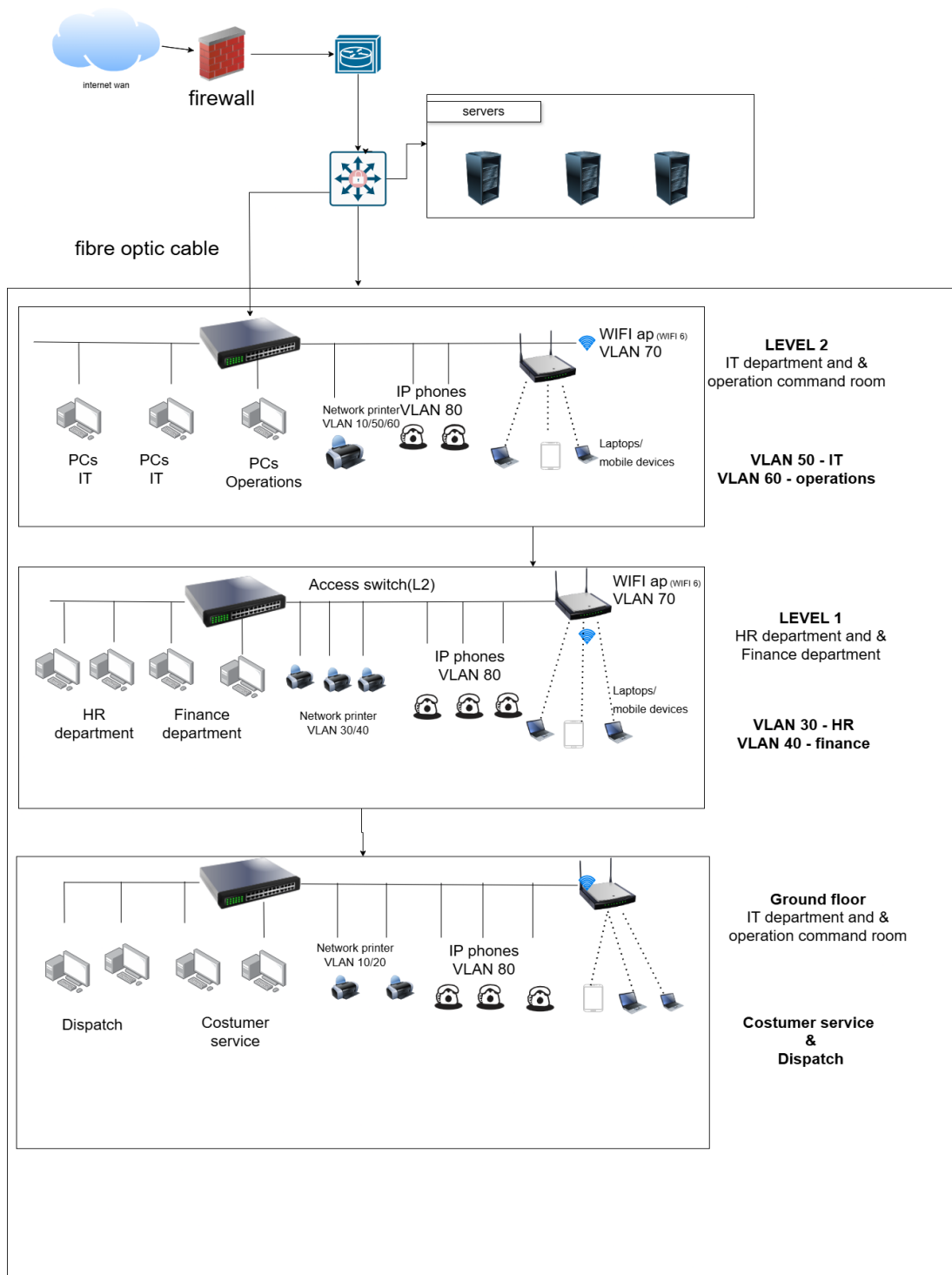


Figure 1: Proposed LAN Design for Sydney Headquarters

LAN Topology, Protocols and Media

Every LAN has a topology describing how the devices in the network are laid out and communicate with one another. The layout and connection of workstations and other LAN devices to one another is referred to as physical topology. But, the way they communicate (i. e. how they send signals to each other) is referred to as logical topology. A LAN can sometimes have different physical and logical topologies.

At the Sydney headquarters, the recommended LAN topology is star topology. With this setup, all network devices get wired through a central switch. Star topology still remains the choice of many companies around the globe as it gives the best combination of reliability, easier fault finding, simplification of management, and scalability that surpass bus or ring network topologies.

In case a device or cable fails, the rest of the network is operational and is not affected. This is why a logistics firm always needs continuous communication and operation, making the topology extremely suitable for them. Every LAN has a topology describing how the devices in the network are laid out and communicate with one another. The layout and connection of workstations and other LAN devices to one another is referred to as physical topology. But, the way they communicate (i. e. how they send signals to each other) is referred to as logical topology. A LAN can sometimes have different physical and logical topologies.

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In case any device malfunctions, the entire system won't crash unlikely to other types of topologies. This is why a logistics firm always needs continuous communication and operation, making the topology extremely suitable for them.



Fig: star topology

Networking protocols

A network protocol is an established set of rules that determine how data is transmitted between different devices in same network. It allows connected devices to communicate with each other, regardless of any differences in their internal processes, structure, or design. Network protocols are the reason you can easily communicate with people all over the world and thus play a significant role in modern day digital communications.

The following networking protocols are used within the Sydney head office **LAN**:

- **TCP/IP (Transmission Control Protocol / Internet Protocol)**
Used as the main communication protocol for transferring data between devices within the network and accessing the internet.
- **DHCP (Dynamic Host Configuration Protocol)**
Automatically assigns IP addresses to computers, printers, phones, and wireless devices connected to the network.
- **DNS (Domain Name System)**
Converts domain names or server names into IP addresses so users can access websites and network resources easily.
- **HTTP/HTTPS (HyperText Transfer Protocol / Secure HyperText Transfer Protocol)**
Used for accessing websites, cloud services, and web-based business applications securely.
- **FTP/SFTP (File Transfer Protocol / Secure File Transfer Protocol)**
Used for transferring files between computers and servers within the organisation.
- **IEEE 802.11 (Wi-Fi Protocols)**
Provides wireless communication between laptops, tablets, barcode scanners, mobile devices, and wireless access points.
- **SIP (Session Initiation Protocol)**
Used to establish and manage VoIP voice and video calls within the organisation.
- **RTP (Real-Time Transport Protocol)**
Used to transmit real-time voice and video data for VoIP communication.
- **VLAN Protocols (IEEE 802.1Q)**
Used to separate departments into different VLANs for better security, traffic management, and network performance.

Communication Media Used in the LAN

The Sydney head office network uses both wired and wireless communication media to support reliable and efficient communication between devices.

Wired Media

- **Cat6 Ethernet Cable**

Within each floor, the Cat6 Ethernet cabling Mostly helps the connection between PCs, printers, VoIP phones, wireless access points, and switches. Cat6 cables are capable of providing high-speed data transfer rates of up to 10 Gbps, although this is feasible only over very short distances. Besides, these cables are an economically viable solution for office setups. They also ensure a very stable and reliable mode of communication, being less prone to signal attenuation.

- **Fiber Optic Cable**

Fiber optic cabling is widely used to connect the core switch and the different floor access switches through the backbone. The fiber optic medium is capable of very high bandwidths, providing fast data transfer and long-distance communication with minimal interference. It is a highly dependable and ideal solution for managing considerable quantities of network traffic between various departments and servers.

Wireless Media

- **Wi-Fi 6 (IEEE 802.11ax)**

Wi-Fi 6 access points installed on each floor providing wireless communication. Compared to older wireless technologies, Wi-Fi 6 not only enhances the speed of wireless communication but also increases the number of devices, better performance, and less delay. It can support the wiring of a laptop tablet barcode scanner, and mobile devices which are used by the staff all over the office.

Justification of Selected Media

Media Type	Reason for Selection
Cat6 Ethernet	Reliable, affordable, high-speed wired communication
Fiber Optic	High bandwidth, low interference, strong backbone performance
Wi-Fi 6	Flexible wireless access, better coverage, supports many devices

The selected communication media provide ABC Logistics with a reliable, scalable, and high-performance network infrastructure suitable for both wired and wireless users.

Internet Access for Head Office

The Sydney HQ will get a fiber business internet connection from an ISP. Fiber internet is an excellent option since it offers lightning-fast speeds, dependable performance, and a consistently stable connection even during complex business transactions (i.e. cloud services), VoIP calls, video conferencing, and inter-office communication.

The internet link is first directed to a router and firewall then the internal network. The router manages the flow of traffic between the LAN and the internet, whereas the firewall is the main gate of the network protecting against unauthorized entries and cyber threats.

This solution is suitable for ABC Logistics because it offers:

- High-speed internet access
- Better security
- Reliable communication
- Support for multiple users and devices
- Easy future expansion

Network operating system

What is the network operating system?

At its core, an operating system simply acts as the liaison between computer hardware and humans. In our everyday lives, when we use devices, a good **GUI** and lots of features come from the operating system installed. Similarly, a network operating system (**NOS**) is a type of software that physically connects different devices and computers in a network so that they can communicate and share resources.

Windows Server is a good option for ABC Logistics as it is a robust platform that **IT** personnel find straightforward to maintain and support, and it is widely adopted by commercial businesses. It makes easier management of users in a centralized manner via Active Directory, This way employees belonging

to different departments such as **HR** Finance Operations, and Customer Service, for instance, can be granted network access after individual account verification.

One of the key benefits for Windows Server is user administration. The IT department can set up individual user accounts for various departments like HR Finance Customer Service, and Operations. Also, different access rights can be given to the employees so that only authorized personnel have access to sensitive company data.

Network Operating Systems (NOS) feature powerful security and authentication capabilities. To block unauthorized personnel from accessing network resources, they require employees to log in with a username and password before gaining access to any network resources. They also strengthen protection through three categories of functions: firewall support, access control, and security policies.

WAN network design

A WAN connects multiple LANs or MANs, enabling data exchange and communication over large distances. This is achieved using routers that link the LANs on both sides of the connection. WANs are essential for organizations with offices in different locations, allowing them to operate as a unified network.

The following diagram illustrates the Wide Area Network (WAN) design proposed for ABC Logistics. The Sydney head office acts as the central network location and is connected to the Brisbane, Canberra, and Adelaide branch offices through the Internet/WAN using secure VPN communication.

The Sydney head office consists of a three-floor building where different departments are connected through switches, routers, wireless access points, and a centralized firewall. The server room located in the IT and Operations floor hosts important company services such as file storage, applications, and network management.

Each branch office contains its own Local Area Network (LAN) with routers, firewalls, switches, PCs, and printers. Routers are used to transfer data between networks and enable communication between offices, while firewalls help protect the company's network from unauthorized access and cyber threats.

The WAN design allows ABC Logistics to securely share files, applications, and resources between all office locations while maintaining reliable communication and centralized management across the organization.

WAN-Diagram of ABC logistic WAN Connectivity Diagram – ABC Logistics

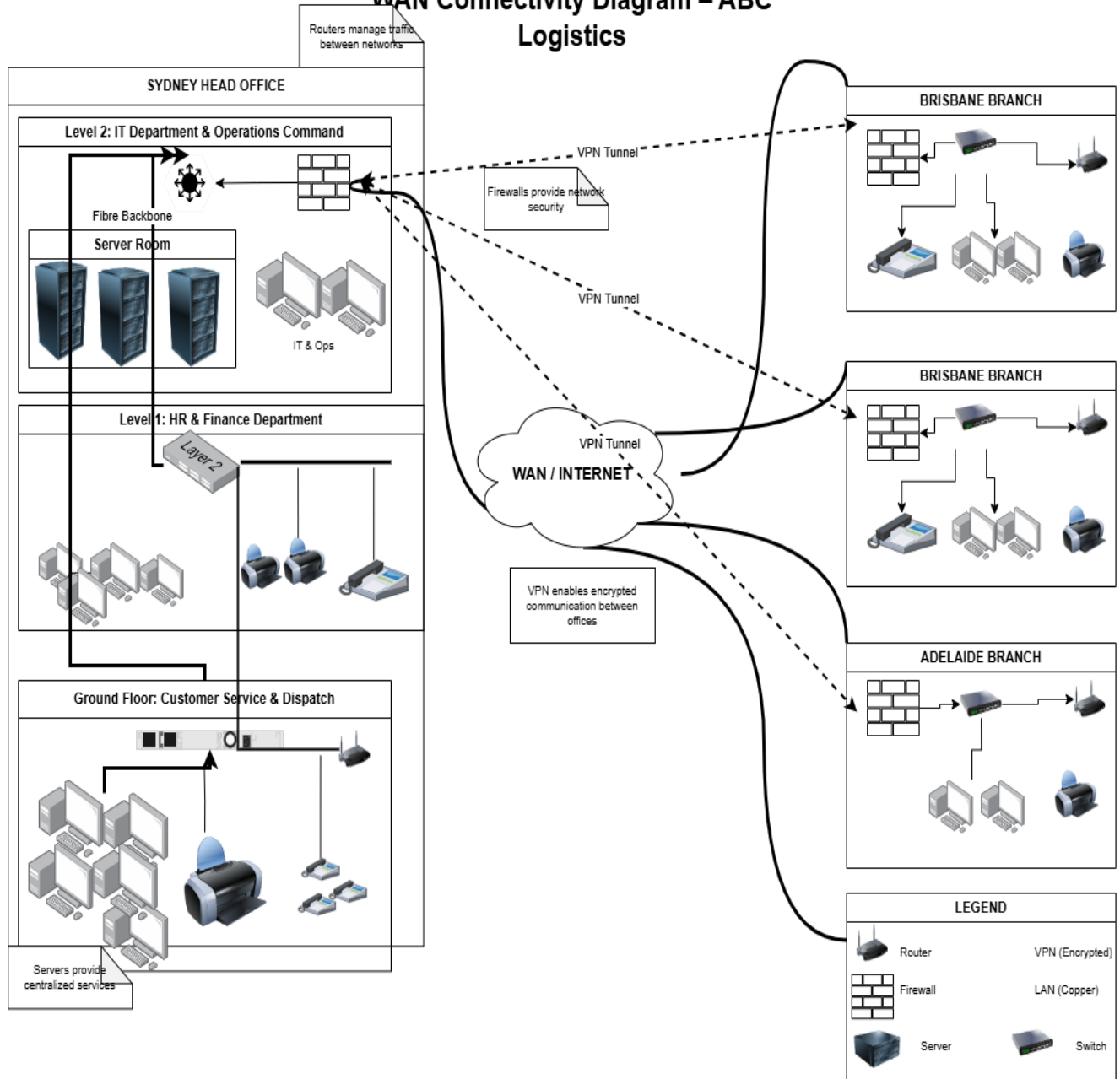


fig: WAN network diagram

WAN Media Selection

ABC Logistics makes use of fibre-based Internet WAN connections in linking to the head office in Sydney with the branches in Brisbane, Canberra, and Adelaide. The offices communicate via VPN connections over the Internet, to ensure secure communication, this type of WAN setup ensures multi-branch organizations achieve integrity, speed, and privacy.

The selection fiber internet as the primary WAN medium is because it provide high bandwidth and rapid data transfer rate compared to traditional copper connections. This feature of the cable is important for ABC Logistics as the company engages in cloud services, file sharing, VoIP communication, video meetings, and centralized applications from different locations. Fiber lines are capable of managing heavy network traffic very well without any significant issues in communications.

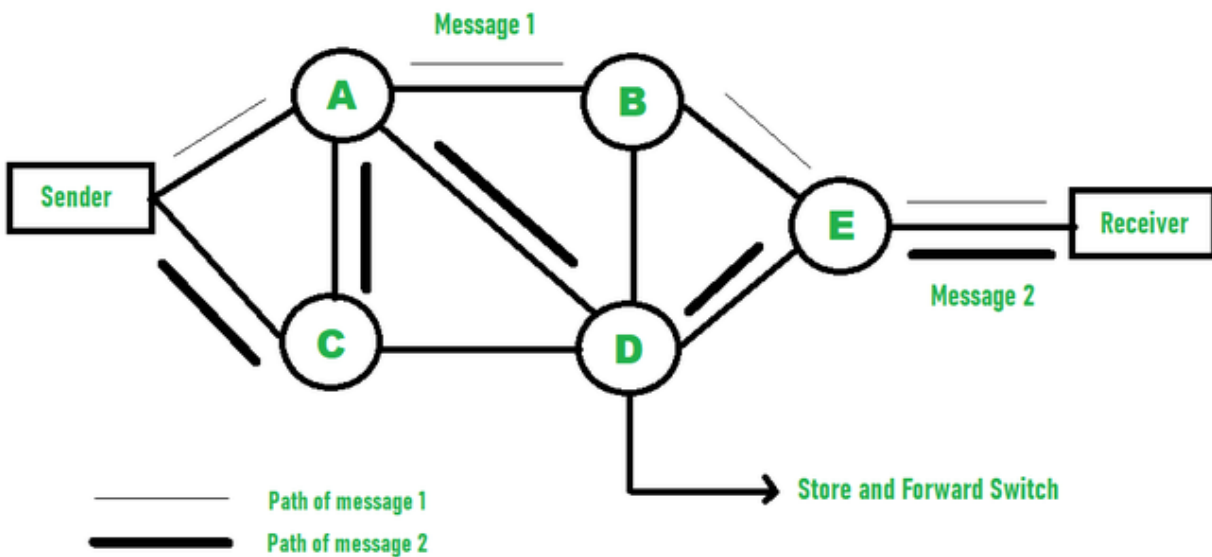
The WAN medium is highly reliable as well. Fiber lines deliver steady communication with very little delay and very little noise, and this results in employees being able to communicate and get company resources without the network being down such easily. Consistent WAN connectivity is very important for everyday logistics operations and communication between branch offices and the head office in Sydney.

Scalability is another important reason for choosing fibre WAN connections, As ABC Logistics grows, number of users, offices devices also grow, and cloud services can be supported more easily without requiring major network changes. The WAN infrastructure can also be upgraded to support higher bandwidth in the future if business requires an increase.

Security is improved using VPN (Virtual Private Network) technology over Internet connection and VPN creates encrypted communication tunnels between the head office and branch offices, helping protect sensitive business information from unauthorized access and cyber threats during data transmission. Hence, reducing vulnerability.

Network Structure and Connection Type

Packet Switching is a method of sending data in small units called packets, each containing a header (control info) and payload. Packets travel independently using the Store-and-Forward technique, possibly via different paths, and are reassembled at the destination. No dedicated path or resource reservation is needed, making data transfer efficient and flexible.



Packet-switched networks are more appropriate than circuit-switched networks for ABC Logistics as it uses cloud services, file sharing, VoIP communication, email services, video meetings, and Internet-based applications. These services require flexible and efficient data transmission rather than a dedicated communication line. Packet switching ensures better scalability and lower operational costs for multi-office companies like ABC logistics.

Circuit-switched networks are mainly designed for traditional telephone systems where a dedicated communication path is established during a call. Although reliable for voice communication, circuit switching is less efficient for modern data communication because network resources remain reserved even when no data is being transferred.

The network mainly uses asynchronous communication services such as emails, cloud access, file transfers, and general Internet usage. Asynchronous communication allows data to be transmitted whenever needed without requiring devices to communicate at perfectly synchronized times. This makes the network more flexible and efficient for everyday business operations.

However, the network also supports isochronous communication for real-time services such as VoIP calls and video conferencing. Isochronous communication helps maintain consistent timing and smooth delivery of voice and video traffic, reducing delays, jitter, and communication interruptions.

Synchronous communication is less suitable for the overall network because most business traffic does not require constant synchronized transmission between devices.

Routing Protocol Recommendation

For the ABC Logistics WAN environment, Open Shortest Path First is a suitable routing protocol recommendation. **OSPF** is used more than RIP in larger scale networks because it has a more efficient system for communication between routers and because it scales better to larger networks. Only changes to the routing table are sent to all the other routers in the network, as opposed to sending the entire routing table at a regular interval (which is how RIP functions).

OSPF is appropriate for ABC Logistics because the company has multiple office locations connected through a WAN. The protocol allows routers at the Sydney head office and branch offices in Brisbane, Canberra, and Adelaide to exchange routing information automatically. This helps the network quickly identify the most efficient communication path between locations.

One major advantage of Open Shortest Path First is routing efficiency. OSPF continuously updates routing tables and can automatically choose alternative paths if a network link fails. This improves reliability and reduces communication downtime between offices.

OSPF is also highly scalable, making it suitable for growing businesses like ABC Logistics. If the company expands by adding more departments, devices, or branch offices in the future, the routing protocol can easily support the larger network without major configuration changes.

Another advantage is faster convergence. When network changes occur, OSPF quickly updates routing information so communication can continue with minimal disruption. This is important for services such as cloud applications, VoIP calls, file sharing, and real-time business communication.

Summary

This report is about a proposed LAN and WAN network design for ABC Logistics. The design focused on creating a reliable and secure network infrastructure to support and communicate between the Sydney head office and the Brisbane, Canberra, and Adelaide branch offices.

Several designs and decisions were made throughout the report, and star topology was chosen for the LAN environment as it provides better performance, easier management, and improved fault isolation

and high data transfer speed. Fibre optic and Cat6 cabling were used to provide high-speed and reliable networks, while wireless Wi-Fi signal is implemented to support wireless devices used by all of the employees.

For WAN environment, a fibre-based Internet connection with VPN communication is used to safely connect all branch offices to the head office. OSPF (open shortest path) was selected as the routing protocol because it provides efficient routing, scalability, and reliable communication between office locations.

The above mentioned solution makes file sharing, cloud access, VoIP communication, wireless connectivity, and centralized management secure and fast across offices. Security is boosted using firewalls, VLANs, authentication(MFAs), and VPNs.

While designing, several risks were considered, including network security, future expansion, communication reliability, bandwidth requirements, and support for both wired and wireless devices. Overall, the proposed network design provides ABC Logistics with a stable and efficient network suitable for modern business operations and future growth.

References

- [1] A. S. Tanenbaum and D. J. Wetherall, *Computer Networks*, 5th ed. Boston, MA, USA: Pearson, 2011.
- [2] J. F. Kurose and K. W. Ross, *Computer Networking: A Top-Down Approach*, 8th ed. New York, NY, USA: Pearson, 2021.
- [3] W. Stallings, *Data and Computer Communications*, 11th ed. London, U.K.: Pearson, 2020.
- [4] [Cisco Systems](#), "Enterprise Network Design Overview," accessed May 2026.
- [5] [Microsoft Learn](#), "Windows Server Documentation," accessed May 2026.
- [6] [IEEE Standards Association](#), "IEEE 802 Networking Standards," accessed May 2026.
- [7] [GeeksforGeeks](#), "Computer Networking Tutorials and Protocols," accessed May 2026.
- [8] [Wikipedia](#), "Computer Network and WAN Technologies," accessed May 2026.